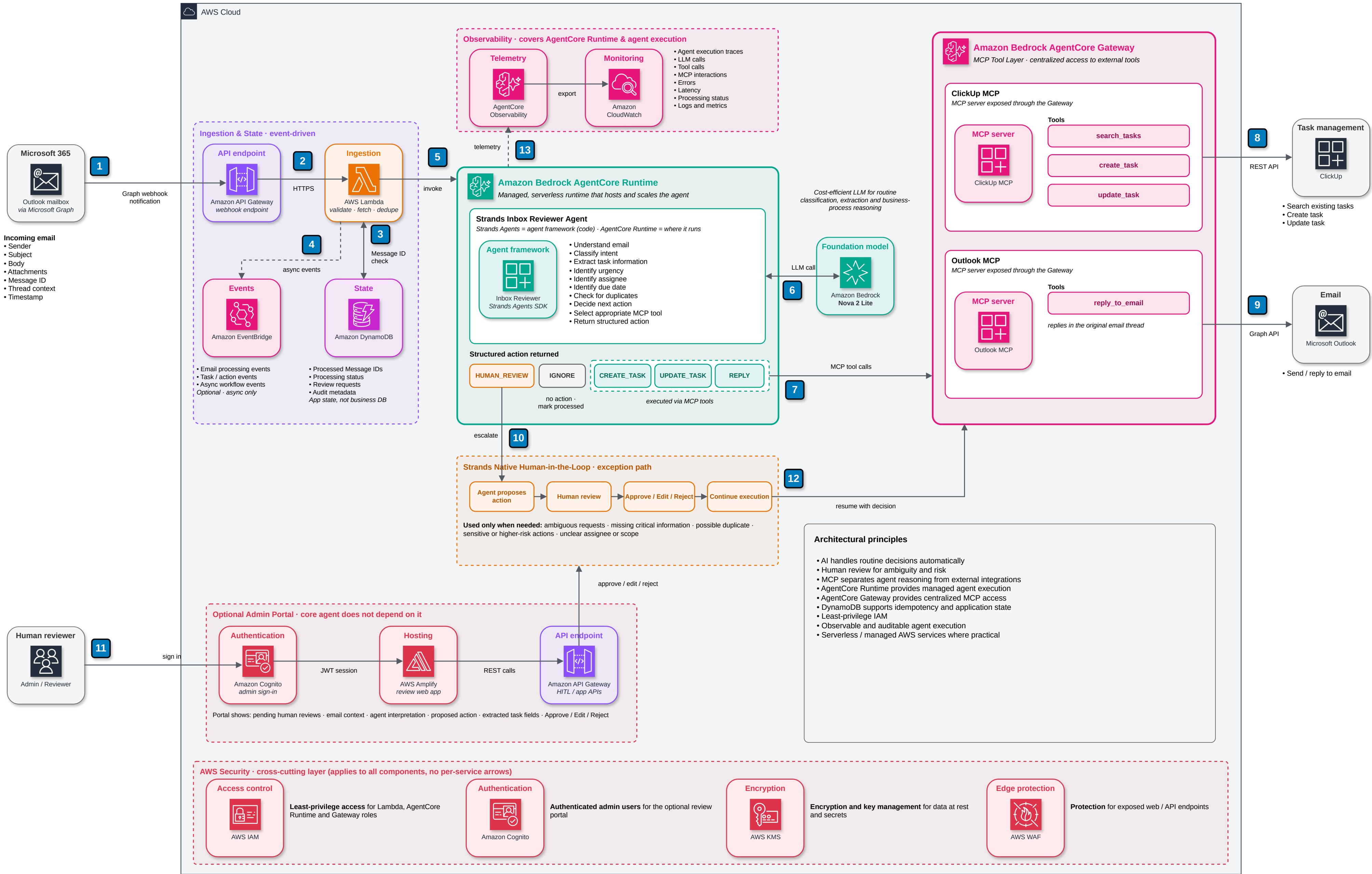


Part 1 – AWS Architecture: AI-Powered Email-to-ClickUp Automation

A Strands agent on Bedrock AgentCore triages Outlook work-request emails: create/update a ClickUp task, reply, ignore, or escalate to human review



- 1 Webhook notification**
Outlook, through Microsoft Graph, pushes a change notification to the API Gateway endpoint. API Gateway only receives it; it never reads the mailbox.
- 2 Ingestion**
API Gateway invokes the ingestion Lambda, which validates the notification and fetches the full email (sender, subject, body, attachments, thread, timestamp).
- 3 Idempotency check**
Lambda checks and records the Message ID in DynamoDB so the same email never creates a duplicate task.
- 4 Async events**
Lambda emits email-processing events to EventBridge (optional asynchronous backbone).
- 5 Invoke agent**
Lambda calls the Strands Inbox Reviewer Agent hosted on AgentCore Runtime.
- 6 LLM reasoning**
The agent calls Amazon Bedrock Nova 2 Lite to classify intent, extract task fields and check for duplicates.
- 7 Act through MCP**
CREATE_TASK, UPDATE_TASK and REPLY are executed as MCP tool calls through AgentCore Gateway. IGNORE takes no action.
- 8 ClickUp tools**
ClickUp MCP exposes search_tasks, create_task and update_task against ClickUp.
- 9 Outlook reply**
Outlook MCP exposes reply_to_email to answer the sender in the original thread.
- 10 Escalate (exception)**
HUMAN_REVIEW is used only for ambiguity, missing information, possible duplicates or higher-risk actions.
- 11 Admin portal (optional)**
A reviewer signs in with Cognito, uses the Amplify app and API Gateway to approve, edit or reject the proposal.
- 12 Resume**
The human decision returns to the agent and execution continues through Gateway and MCP.
- 13 Observability**
AgentCore Observability sends traces, LLM, tool and MCP calls, latency and errors to CloudWatch.

Line styles
Solid arrow: synchronous call / data flow
Dashed arrow: asynchronous events and telemetry
Orange container: human-in-the-loop exception path
Red dashed container: optional admin portal